ANSWERS

1 **(a)** 2.4 s .... A1 [1]

(b) in **(b)** and **(c)**, allow answers as (+) or (–)

recognises distance travelled as area under graph line C1

height = (½ × 2.4 × 9.0) – (½ × 1.6 × 6.0) C1

= 6.0 m (*allow 6 m*) .... A1 [3]

(*answer 15.6 scores 2 marks*

answer 10.8 or 4.8 scores 1 mark)

alternative solution: $s = ut - ½at^2$

$= (9 × 4) - ½ × (9 / 2.4) × 4^2$

$= 6.0$ m

(*answer 66 scores 2 marks*

answer 36 or 30 scores 1 mark)

(c) **(i)** change in momentum = 0.78 (9.0 + 4.2) (*allow 4.2 ± 0.2*) C1

= 10.3 N s (allow 10 N s) .... A1 [2]

(ii) force = $\Delta p / \Delta t$ or $m\Delta v / \Delta t$ C1

= 10.3 / 3.5 / 0.08

= 2.9 N .... A1 [2]

(d) **(i)** 2.9 N .... A1 [1]

(ii) g = weight / mass C1

= 2.9 / 0.78

= 3.7 m s^{-2} .... A1 [2]

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2 **(a)** **(i)** *either* sum / total momentum (of system of bodies) is constant

or <u>total</u> momentum before = <u>total</u> momentum after M1

for an isolated system / no (external) force acts on system .... A1 [2]

(ii) zero momentum before / after decay M1

so α-particle and nucleus D must have <u>momenta</u> in opposite directions .... A1 [2]

(b) **(i)** kinetic energy = ½ mv^2 .. ..C1

$1.0 \times 10^{-12} = ½ \times 4 \times \underline{1.66} \times 10^{-27} \times v^2$ M1

$v = 1.7 \times 10^7$ m s^{-1} .... A0 [2]

(ii) $1.7 \times 10^7 \times 4u = 216u \times V$ C1

$V = 3.1 \times 10^5$ m s^{-1} .... A1 [2]

(*accept* 3.2×10^5 *m* s^{-1}*, do not accept 220 rather than 216*)

(c) $(1.7 \times 10^7)^2 = 2 \times$ *deceleration* $\times 4.5 \times 10^{-2}$ C1

deceleration / a = 3.2×10^{15} m s^{-2} .... A1 [2]

(*accept calculation based on calculating* $F = 2.22 \times 10^{-11}$ *N*

and then use of $F = ma$)

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3 **(a)** **(i)** force is rate of change of momentum ………………………………………… B1 [1]
(ii) force on body A is equal in magnitude to force on body B (from A) …………M1
forces are in opposite directions …………………………………………… A1
forces are of the same kind ……………………………………………………A1 [3]
(b) **(i)** 1 $F_A = -F_B$ …………………………………………………………… B1 [1]
2 $t_A = t_B$ …………………………………………………………… B1 [1]
(ii) $\Delta p = F_A t_A = -F_B t_B$ ………………………………………………….. B1 [1]
(c) graph: momentum change occurs at same times for both spheres ………… B1
final momentum of sphere B is to the right ……………………………….. M1
and of magnitude 5 N s …………………………………………………… A1 [3]

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4 **(a)** force = rate of change of momentum (allow symbols if defined) B1 [1]
(b) **(i)** $\Delta\rho = 140 \times 10^{-3} \times (5.5 + 4.0)$ C1
$= 1.33\ \text{kg m s}^{-1}$ A1 [2]
(ii) force = 1.33 / 0.04 M1
= 33.3 N A0 [1]
(c) **(i)** taking moments about B C1
$(33 \times 75) + (0.45 \times g \times 25) = F_A \times 20$ C1
$F_A = 129$ N A1 [3]
(ii) $F_B = 33 + 129 + 0.45g$ C1
= 166 N A1 [2]

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5 **(a)** (resultant) force = rate of change of momentum / allow proportional to
or change in momentum / time (taken) B1 [1]
(b) **(i)** $\Delta p = (-)\ 65 \times 10^{-3}\ (5.2 + 3.7)$ C1
= (–) 0.58 N s A1 [2]
(ii) $F = 0.58 / 7.5 \times 10^{-3}$
= 77(.3) N A1 [1]
(c) **(i)** **1.** force on the wall from the ball is equal to the force on ball from the wall M1
but in the opposite direction A1 [2]
(statement of Newton's third law can score one mark)
2. momentum change of ball is equal and opposite to momentum change
of the wall / change of momentum of ball and wall is zero B1 [1]
(ii) kinetic energy (of ball and wall) is reduced / not conserved so inelastic B1 [1]
(Allow relative speed of approach does not equal relative speed of separation.)

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6 **(a)** force = rate of change of momentum A1 [1]
(b) **(i)** horizontal line on graph from $t = 0$ to t about 2.0 s ± ½ square, $a > 0$ M1
horizontal line at 3.5 on graph from 0 to 2 s A1
vertical line at $t = 2.0$ s to $a = 0$ or sharp step without a line B1
horizontal line from $t = 2$ s to $t = 4$ s with $a = 0$ B1 [4]
(ii) straight line and positive gradient M1
starting at (0,0) A1
finishing at (2,16.8) A1
horizontal line from 16.8 M1
from 2.0 to 4.0 A1 [5]

7	**(i)**	the total momentum of a system (of interacting bodies) remains constant	M1	
		provided there are no resultant external forces / isolated system	A1	[2]
	(ii)	elastic: total kinetic energy is conserved, inelastic: loss of kinetic energy	B1	[1]
		[allow elastic: relative speed of approach equals relative speed of separation]		
(b)	**(i)**	initial mom: 4.2 × 3.6 – 1.2 × 1.5 (= 15.12 – 1.8 = 13.3)	C1	
		final mom: 4.2 × v + 1.5 × 3	C1	
		v = (13.3 – 4.5) / 4.2 = 2.1 m s^{-1}	A1	[3]
	(ii)	initial kinetic energy = ½ $m_A(v_A)^2$ + ½ $m_B(v_B)^2$		
		= 27.21 + 1.08 = 28(.28)	M1	
		final kinetic energy = 9.26 + 6.75 = 16	M1	
		initial KE is not the same as final KE hence inelastic	A1	[3]
		provided final KE less than initial KE		
		[allow in terms of relative speeds of approach and separation]		

8	**(a)**	a body / mass / object continues (at rest or) at constant / uniform velocity unless acted on by a <u>resultant</u> force	B1	[1]
(b)	**(i)**	weight <u>vertically</u> down	B1	
		normal / reaction / contact (force) perpendicular / normal <u>to the slope</u>	B1	[2]
	(ii) 1.	acceleration = gradient *or* $(v - u)/t$ *or* $\Delta v/t$	C1	
		= (6.0 – 0.8) / (2.0 – 0.0) = 2.6 m s^{-2}	M1	[2]
	2.	$F = ma$		
		= 65 × 2.6		
		= 169 N (allow to 2 or 3 s.f.)	A1	[1]
	3.	weight component seen: mg sinθ (218 N)	C1	
		218 – R = 169	C1	
		R = 49 N (require 2 s.f.)	A1	[3]

9	**(a)**	for a system (of interacting bodies) the <u>total</u> momentum remains constant	M1	
		provided there is no <u>resultant</u> force acting (on the system)	A1	[2]
(b)	**(i)**	total momentum = $m_1v_1 + m_2v_2$	C1	
		= 0.4 × 0.65 + 0.6 × 0.45	C1	
		= 0.26 + 0.27 = 0.53 N s	A1	[3]
	(ii)	0.53 = 0.4 × 0.41 + 0.6 × v	C1	
		v = 0.366 / 0.6 = 0.61 m s^{-1}	A1	[2]
	(iii)	KE = ½mv^2	C1	
		total initial KE = ½ × 0.4 × $(0.65)^2$ + ½ × 0.6 × $(0.45)^2$	C1	
		= 0.0845 + 0.06075 = 0.15 (0.145) J	A1	[3]
(c)		check relative speed of approach equals relative speed of separation *or:* total final kinetic energy equals the total initial kinetic energy	B1	[1]
(d)		the forces on the two bodies (or on X and Y) are equal and opposite	B1	
		time same for both forces <u>and</u> force is change in momentum / time	B1	[2]

10	**(a)** $4.5 \times 50 - 2.8 \times M$ (= ...)	C1	
	(...) = $-1.8 \times 50 + 1.4 \times M$	C1	
	(M =) 75 g	A1	[3]
	(b) <u>total</u> initial kinetic energy/KE not equal to the total final kinetic energy/KE		
	or <u>relative</u> speed of approach is not equal to relative speed of separation		
	so not elastic or is inelastic	B1	[1]
	(c) force on X is equal and opposite to force on Y (Newton III)	M1	
	force equals/is proportional to rate of change of momentum (Newton II)	M1	
	time of collision same for both balls hence change in momentum is the same	A1	[3]
11	**(a)** (p =) mv	C1	
	Δp (= $-6.64 \times 10^{-27} \times 1250 - 6.64 \times 10^{-27} \times 1250$) = 1.66×10^{-23} N s	A1	[2]
	(b) **(i)** molecule collides with wall/container **and** there is a change in momentum	B1	
	change in momentum / time is force or $\Delta p = Ft$	B1	
	<u>many/all/sum of</u> molecular collisions over surface/area of container produces pressure	B1	[3]
	(ii) more collisions per unit time so greater pressure	B1	[1]
12	**(a)** **(i)** force (= $mg = 0.15 \times 9.81$) = 1.5 (1.47) N	A1	[1]
	(ii) resultant force (on ball) is zero so normal contact force = weight		
	or the forces are in opposite directions so normal contact force = weight		
	or normal contact force up = weight down	A1	[1]
	(b) **(i)** (resultant) force proportional/equal to rate of change of momentum	B1	[1]
	(ii) change in momentum = $0.15 \times (6.2 + 2.5)$ (= 1.305 N s)	C1	
	magnitude of force = 1.305 / 0.12		
	= 11 (10.9) N	A1	
	or		
	(average) acceleration = (6.2 + 2.5) / 0.12 (= 72.5 m s^{-2})	(C1)	
	magnitude of force = 0.15×72.5		
	= 11 (10.9) N	(A1)	
	(direction of force is) upwards/up	B1	[3]
	(iii) there is a change/gain in momentum of the floor	M1	
	this is equal (and opposite) to the change/loss in momentum of the ball so momentum is conserved	A1	[2]
	or		
	change of (total) momentum of <u>ball and floor</u> is zero	(M1)	
	so momentum is conserved	(A1)	
	or		
	(total) momentum of <u>ball and floor</u> before is equal to the (total) momentum of <u>ball and floor</u> after	(M1)	
	so momentum is conserved	(A1)	

13	**(a)**	the <u>total</u> momentum of a system (of colliding particles) remains constant	M1	
		provided there is no resultant external force acting on the system/ isolated or closed system	A1	[2]
	(b) (i)	the <u>total</u> kinetic energy before (the collision) is equal to the total kinetic energy after (the collision)	B1	[1]
	(ii)	p (= mv = 1.67 × 10^{-27} × 500) = 8.4 (8.35) × 10^{-25} N s	A1	[1]
	(iii)	**1.** $mv_A \cos 60° + mv_B \cos 30°$ or $m(v_A^2 + v_B^2)^{1/2}$	B1	
		2. $mv_A \sin 60° + mv_B \sin 30°$	B1	[2]
	(iv)	8.35 × 10^{-25} or $500m = mv_A \cos 60° + mv_B \cos 30°$ *and* $0 = mv_A \sin 60° + mv_B \sin 30°$ *or* using a vector triangle	C1	
		v_A = 250 m s^{-1}	A1	
		v_B = 430 (433) m s^{-1}	A1	[3]
14	**(i)**	$a = (v - u) / t$ **or** gradient **or** $\Delta v / (\Delta)t$	**C1**	
		e.g. a = (8.8 – 4.6) / (7.0 – 4.0) = 1.4 m s^{-2}	**A1**	
	(ii)	s = 4.6 × 4 + [(8.8 + 4.6) / 2] × 3	**C1**	
		= 18.4 + 20.1	**A1**	
		= 39 (38.5) m		
	(iii)	ΔE = ½ × 95 [$(8.8)^2 - (4.6)^2$]	**C1**	
		= 3678 – 1005	**A1**	
		= 2700 (2673) J		
	(iv) 1	weight = 95 × 9.81 (= 932 N)	**C1**	
		vertical tension force = 280 sin 25° **or** 280 cos 65° (=118.3 N)	**C1**	
		F = 932 + 118	**A1**	
		= 1100 (1050) N		
	(iv) 2	horizontal tension force = 280 cos 25° **or** 280 sin 65° (= 253.8 N)	**C1**	
		resultant force = 95 × 1.4 (= 133 N)	**C1**	
		133 = 253.8 – R	**A1**	
		R = 120 (120.8) N		

15 a body/mass/object continues (at rest or) at constant/uniform velocity unless acted on by a resultant force

(b) (i) initial momentum = final momentum **C1**

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$0.60 \times 100 - 0.80 \times 200 = -0.40 \times 100 + v \times 200$ **A1**

$v = (-)\ 0.3(0)\ \text{m s}^{-1}$

(ii) kinetic energy is not conserved/is lost (but) total energy is conserved/constant **B1**

or some of the (initial) kinetic energy is transformed into other forms of energy **C1**

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16 (a) (i) $K\rho = 1.5 / 33$

$= 1.38 \times 10^{-3}$ **A1**

$F_D = 1.38 \times 10^{-3} \times 25^2$ **or** $F_D / 1.5 = 25^2 / 33^2$

$F_D = 0.86\ \text{N}$

(ii) $a = (1.5 - 0.86) / (1.5 / 9.81)$ **or** $a = 9.81 - [0.86 / (1.5 / 9.8$ **C1**

$a = 4.2\ \text{m s}^{-2}$ **A1**

(b) initial acceleration is g/9.81 (m s^{-2})/acceleration of free fall **B1**

acceleration decreases **B1**

final acceleration is zero **B1**

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17 (a) sum/total momentum (of system of bodies) is constant **M1**

or sum/total momentum before = sum/total momentum after

for an isolated system/no (resultant) external force **A1**

(b) (i) $p = mv$ **C1**

$(4.0 \times 6.0 \times \sin\theta) - (12 \times 3.5 \times \sin 30°) = 0$ **M1**

or

$(m_Av_A \times \sin\theta) - (m_Bv_B \times \sin 30°) = 0$

$\theta = 61°$ **A1**

(ii) shows the horizontal momentum component of ball A or of ball B as $(4.0 \times 6.0 \times \cos\theta)$ **or** $(12 \times 3.5 \times \cos 30°)$ **C1**

$(4.0 \times 6.0 \times \cos 61°) + (12 \times 3.5 \times \cos 30°) = 4.0v$ so $v = 12\ (\text{m s}^{-1})$ **A1**

(iii) initial E_K (= ½ × 4.0 × 12^2) = 290 (288) (J) **M1**

final E_K (= ½ × 4.0 × 6.0^2 + ½ × 12 × 3.5^2) = 150 (145.5) (J) **M1**

(initial E_K > final E_K) so inelastic [both M1 marks required to award this mark] **A1**

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18 **(a)** mass is the property (of a body/object) resisting changes in motion **B1**
or mass is the quantity of matter (in a body)

(b) **(i)** force on A (by B) equal and opposite to force on B (by A) **or**
both A and B exert equal and opposite forces on each other **B1**
force is rate of change of momentum and time (of contact) is same **B1**

(ii) $p = mv$ **or** $3M \times 0.40$ **or** $M \times 0.25$ **or** $3M \times 0.2$ **or** Mv **C1**
$(3M \times 0.40) - (M \times 0.25) = (3M \times 0.2) + Mv$ **C1**
$v = (3 \times 0.40) - 0.25 - (3 \times 0.2)$ **A1**
$= 0.35\,\text{m s}^{-1}$

(iii) **1.** relative speed of approach $= 0.40 + 0.25$ **A1**
$= 0.65\,\text{m s}^{-1}$

2. relative speed of separation $= 0.35 - 0.20$ **A1**
$= 0.15\,\text{m s}^{-1}$

(iv) (relative) speed of separation not equal to/less than (relative) speed of approach **or** answers (to (b)(iii) are) not equal **and** so inelastic collision **B1**

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19 **(a)** sum/total momentum (of a system of bodies) is constant **M1**
or sum/total momentum before = sum/total momentum after

for an isolated system **or** no (resultant) external force **A1**

(b) **(i)** (p =) mv **or** $(3.0M \times 7.0)$ **or** $(2.0M \times 6.0)$ **or** $(1.5M \times 8.0)$ **C1**
$3.0M \times 7.0 = 2.0M \times 6.0\sin\theta + 1.5M \times 8.0\sin\theta$ **C1**
$\theta = 61°$ **A1**

or (vector triangle method)
momentum vector triangle drawn **(C1)**
$\theta = 61°$ (*2 marks for* $\pm1°$, *1 mark for* $\pm2°$) **(A2)**

or (use of cosine rule)
$p = mv$ **or** $(3.0M \times 7.0)$ **or** $(2.0M \times 6.0)$ **or** $(1.5M \times 8.0)$ **(C1)**
$(21M)^2 = (12M)^2 + (12M)^2 - (2 \times 12M \times 12M \times \cos 2\theta)$ **(C1)**
$\theta = 61°$ **(A1)**

(ii) (E =) $\frac{1}{2}mv^2$ **C1**

ratio $= (\frac{1}{2} \times 2.0M \times 6.0^2) / (\frac{1}{2} \times 1.5M \times 8.0^2)$ **A1**
$= 0.75$

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20 (a)	(p =) mv **or** 4.0×45 **or** 2.0×85 **or** $89v$	**C1**
	$(4.0 \times 45) - (2.0 \times 85) = 89\,v$	**A1**
	$v = 0.11\,\mathrm{m\,s^{-1}}$	
(b) (i)	**1.** speed of approach = $47\,\mathrm{m\,s^{-1}}$	**A1**
	and	
	2. speed of separation = 0	
(ii)	speed of separation less than/not equal to speed of approach **and** so inelastic collision	**A1**
(c)	force is equal to rate of change of momentum	**B1**
	force on ball (by block) equal and opposite to force on block (by ball) so rates of change of momentum are equal and opposite	**B1**
	or	
	force on ball (by block) equal and opposite to force on block (by ball)	**(B1)**
	force is equal to rate of change of momentum so rates of change of momentum are equal and opposite	**(B1)**

21 (a)	sum/total momentum (of a system of bodies) is constant	**M1**
	or	
	sum/total momentum before = sum/total momentum after	
	for an isolated system **or** no (resultant) external force	**A1**
(b) (i)	$m = \rho V$	**C1**
	$= 1.3 \times \pi \times 0.045^2 \times 1.8 \times 2.0 = 0.030$ (kg)	**A1**
(ii)	**1.** $(\Delta)p = (\Delta)mv$	**C1**
	$= 0.030 \times 1.8$	**A1**
	$= 0.054\,\mathrm{N\,s}$	
	2. $F = 0.054/2.0$ **or** $0.030 \times 1.8/2.0$	**A1**
	$= 0.027\,\mathrm{N}$	
(iii)	force on air (by propeller) equal to force on propeller (by air)	**M1**
	and opposite (in direction)	**A1**
(iv)	resultant force $= 0.20 \times 0.075$ (= 0.015 N)	**C1**
	frictional force $= 0.027 - 0.015$	
	$= 0.012\,\mathrm{N}$	**A1**

22	**(a)**	(resultant) force proportional/equal to rate of change of momentum	**B1**
	(b)(i)	$\rho = m / V$	**C1**
		$V = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2$ (= 4.59×10^{-4} m^3)	**A1**
		$m = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2 \times 1000 = 0.46$ kg	
	(ii)	**1.** $(\Delta)p = (\Delta m)v$	**C1**
		$(\Delta)p = 0.46 \times 13$	**A1**
		= 6.0 N s	
		2. $F = 6.0 / 0.20$	**A1**
		= 30 N	
	(iii)	force on water (by rocket/nozzle) equal to force on rocket/nozzle (by water)	**M1**
		in the opposite direction	**A1**
	(iv)	**1.** mass = 0.40 + 0.70 – 0.46 = 0.64 kg	**A1**
		2. acceleration = $[30 - (0.64 \times 9.81)] / 0.64$ **or** $30 / 0.64 - 9.81$	**C1**
		= 37 m s^{-2}	**A1**

23	**(a)**	(momentum =) mass × velocity	**B1**
	(b) (i)	time = 40 ms	**A1**
	(ii)	**1.** (the magnitude of the acceleration is) constant	**B1**
		2. (the magnitude of the acceleration is) zero	**B1**
	(c)	$F = \Delta p / (\Delta)t$ **or** F = gradient	**C1**
		e.g. $F = 0.50 / 40 \times 10^{-3}$	**A1**
		= 13 N	
	(d)	horizontal line from (0, 0.40) to (60, 0.40)	**B1**
		straight line from (60, 0.40) to (100, –0.10)	**B1**
		horizontal line from (100, –0.10) to (160, –0.10)	**B1**

24	**(a)**	(resultant) force proportional to rate of change of momentum	**B1**
	(b)(i)	arrow drawn vertically downwards from point X	**B1**
	(ii)	$s = ut + ½at^2$	**C1**
		$h = ½ \times 9.81 \times 0.81^2$	
		= 3.2 m	**A1**
	(iii)	$d = 5.4 \times 0.81$	**A1**
		= 4.4 m	
	(c)(i)	downward pointing arrow labelled weight	**B1**
		upward pointing arrow labelled air resistance	**B1**
	(ii)	air resistance increases	**B1**
		weight constant **or** <u>resultant</u> force decreases	**B1**
		(so) acceleration decreases	**B1**
	(iii)	gravitational potential energy to thermal/internal energy	**B1**

25 (a) (i) area = $ut + ½(v – u)t$ **A1**
or
area = $vt – ½(v – u)t$
or
area = $½(u + v)t$

(ii) displacement **A1**

(b) (i) u = 15 sin 60° (= 13 m s^{-1}) **C1**
t = 15 sin 60° / 9.81 **C1**
= 1.3 s **A1**

(ii) the force in the horizontal direction is zero **B1**

(iii) (velocity =) 15 cos 60° = 7.5 (m s^{-1}) **A1**
or
(velocity =) 15 sin 30° = 7.5 (m s^{-1})

(c) (i) $p = mv$ **or** 0.40 × 7.5 **or** 0.40 × 4.3 **C1**
Δp = 0.40 (7.5 + 4.3) **A1**
= 4.7 kg m s^{-1}

(ii) force = 4.7 / 0.12 **or** 0.40 × [(7.5 + 4.3) / 0.12] **A1**

= 39 N

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26 (a) a body continues at (rest or) constant velocity unless acted upon by a resultant force **B1**

(b) (i) distance = [½ × (2.0 + 4.4) × 3.0] + [4.4 × 2.0] **C1**
= 9.6 + 8.8 **A1**
= 18 m

(ii) $a = (v – u) / t$ **or** gradient **or** $\Delta v / (\Delta)t$ **C1**
e.g. a = (4.4 – 2.0) / 3.0 = 0.80 m s^{-2} **A1**

(iii) 1. force = 240 cos 28° **or** 240 sin 62° **A1**
= 210 N

2. resultant force = 89 × 0.80 (= 71.2 N) **C1**
R = 210 – 71 **A1**
= 140 N

(iv) T sin 45° = mg **C1**
T = (89 × 9.81) / sin 45° **A1**
= 1200 N

27.

(a)	(force =) rate of change of momentum
(b)(i)	$E = ½mv^2$ **or** $½ \times 0.062 \times 3.8^2$ **or** $½ \times 0.062 \times 1.7^2$
	loss of KE $= ½ \times 0.062 \times (3.8^2 – 1.7^2)$ $= 0.36$ J
(b)(ii)	$p = mv$ **or** 0.062×3.8 **or** 0.062×1.7
	change in momentum $= 0.062 \times (1.7 + 3.8)$ $= 0.34$ N s
b)(iii)	(average resultant force =) $0.34 / 0.081 = 4.2$ (N) **or** (average resultant force =) $0.062 \times (1.7 + 3.8) / 0.081 = 4.2$ (N)
b)(iv)	**1.** average force $= 4.2 + (0.062 \times 9.81)$ $= 4.8$ N
	2. average force $= 4.8$ N

28.

(a)(i)	$(\Delta)p = \rho g(\Delta)h$ $520 = 1000 \times 9.81 \times h$
	$h = 0.053$ m
(a)(ii)	(upthrust =) $(\Delta)p \times A$ $= (\Delta)p \times \pi(d / 2)^2$ **or** $(\Delta)p \times \pi r^2$
	$= 520 \times \pi(0.031 / 2)^2 = 0.39$ (N)
(a)(iii)	$T = 0.84 – 0.39$ $= 0.45$ N
(b)(i)	$a = (v – u) / t$ **or** $(\Delta)v / (\Delta)t$ **or** gradient
	= e.g. $8.0 \times 10^{-2} / 2.0$ $= 4.0 \times 10^{-2}$ m s^{-2}
(b)(ii)	distance $= (½ \times 2.5 \times 0.10) + (½ \times 1.5 \times 0.10)$ **or** $(½ \times 4.0 \times 0.10)$ (= 0.20 (m))
	depth $= 0.32 – 0.20$ $= 0.12$ m
(c)(i)	viscous (force)
(c)(ii)	viscous force increases (with speed/time/depth)
	(so) acceleration decreases

29.

(a)	acceleration: vector electrical resistance: scalar momentum: vector *1 mark for two correct, 2 marks for all three correct*
(b)	resultant force (in any direction) is zero
	resultant torque/moment (about any point) is zero
(c)(i)	upthrust = $\rho g(\Delta)h \times A$
	$= (1.00 \times 10^3 \times 9.81 \times 0.190) \times 0.0230$ $= 42.9$ N
(c)(ii)	(T =) 43 – 28 = 15 (N) **or** (T =) 42.9 – 28 = 14.9 or 15 (N)
(c)(iii)	$\sigma = F / A$ **or** T / A
	$= 15 / (3.2 \times 10^{-6})$ $= 4.7 \times 10^6$ Pa
(c)(iv)	upthrust (on cylinder) increases (and weight constant)
	tension/stress increases and (so) strain energy increases

30.

(a)	$v^2 = u^2 + 2as$ $u^2 = 8.7^2 - (2 \times 9.81 \times 1.5)$
	$u = 6.8$ m s^{-1}
(b)	(magnitude of) force on ball (by ground) equal to force on ground (by ball)
	(direction of) force on ball (by ground) opposite to force on ground (by ball)
(c)(i)	(p =) 0.059×8.7 **or** 0.059×5.4
	change in momentum = 0.059 (8.7 + 5.4) = 0.83 N s
(c)(ii)	resultant force = 0.83 / 0.091 **or** 0.059 [(8.7 + 5.4) / 0.091] = 9.1 N
(c)(iii)	(W =) 0.059×9.81
	(W =) 0.58 (N) force = 9.1 + 0.58 = 9.7 N

(d)	straight line with a positive gradient and starting from a non-zero value of speed at $t = 0$ and ending when $t = T$
(e)	air resistance increases
	resultant force/acceleration decreases so gradient (of curve) decreases

31.

(a)	mass × velocity
(b)(i)	kinetic energy $= \frac{1}{2}mv^2$
	$= \frac{1}{2} \times 0.24 \times 2.3^2$
	$= 0.63$ J
(b)(ii)	change in momentum $= \frac{1}{2} \times 240 \times 5.0 \times 10^{-3}$
	$= 0.60$ N s
(b)(iii)	(change in velocity of Y) $= 0.60 / 0.12$ $(= 5.0\ \text{m s}^{-1})$
	final velocity of Y $= 5.0 – 2.3$ $= 2.7\ \text{m s}^{-1}$
	or
	(final momentum of Y) $= 0.60 – 0.12 \times 2.3$ $(= 0.324$ N s)
	final velocity of Y $= 0.324 / 0.12$ $= 2.7\ \text{m s}^{-1}$
(c)	sloping straight line from (0, 0) to $t = 3.0$ ms **and** another straight line continuous with the first from $t = 3.0$ ms to (5.0, 0)
	lines showing maximum force of magnitude 240 N
	lines wholly in the negative F region of the graph

32.

(a)	Fv: kg m s^{-2} k: kg m s^{-2} / m × m s^{-1}	
	= kg m^{-1} s^{-1}	
(b)	$F = \rho gV$ $V = 4 / 3 \times \pi \times (2.1 \times 10^{-3})^3$	$(=3.88 \times 10^{-8}\ \text{m}^3)$
	$\rho = 4.8 \times 10^{-4} / 9.81 \times V$	
	$= 1300\ \text{kg m}^{-3}$	
(c)(i)	W downwards, U upwards, Fv upwards	
(c)(ii)	$F_V = 7.2 \times 10^{-4} – 4.8 \times 10^{-4}$ $= 2.4 \times 10^{-4}$ (N)	
	velocity $= 2.4 \times 10^{-4} / (17 \times 2.1 \times 10^{-3})$ $= 6.7 \times 10^{-3}\ \text{m s}^{-1}$	

33.

(a)	$(m =)\ \rho V$
	$= 1.0 \times 10^3 \times 1.5 \times 10^{-4} \times 5.0 \times 1.6 = 1.2$ (kg)
(b)(i)	$(\Delta)p = 1.2 \times 5.0$ $= 6.0$ N s
(b)(ii)	$F = 6.0 / 1.6$ or $1.2 \times 5.0/1.6$ $= 3.8$ N
(c)	Newton's third law applies (so) 3.8 N.
(d)	$p = F / A$ $= 3.8 / 1.5 \times 10^{-4}$
	$= 2.5 \times 10^4$ Pa

34.

(a)	change in displacement / time (taken)
(b)(i)	(displacement =) area under graph
	(at $t = 4.0$ s) $v = (–)\ 2.4$
	height $= ½ \times 2.5 \times 4.0 – ½ \times 1.5 \times 2.4$ $= 3.2$ m
(b)(ii)	change in momentum = 7.5 (–4.0 – 2.4)
	= (–) 48 N s
(b)(iii)	$W = \Delta p / (\Delta)t$ **or** $\Delta mv / (\Delta)t$
	$= 48 / 4.0$ $= 12$ N
	or
	$W = ma$ **or** mg **or** $m(v – u) / t$
	$= 7.5 \times 1.6$ **or** $7.5 \times (4 + 2.4) / 4.0$ $= 12$ N
(c)	speed/velocity decreases so viscous force decreases
	viscous force decreases (and weight constant) so resultant force decreases

35.

(a)	$T \sin 68° + 32 = 280$
	$T = 270$ N
(b)(i)	$F = \rho g V$ $V = 280 / (1.0 \times 10^3 \times 9.81)$ $= 0.029$ m^3
(b)(ii)	$\rho = (32 / 9.81) / 0.029$
	$= 110$ kg m^{-3}
(c)	$(\Delta)E = mg(\Delta)h$ **or** $(\Delta)E = W(\Delta)h$
	$(\Delta)h = (-)\ 77 / 32$
	$(\Delta)h = (-)\ 2.4$ final height $= 6.2 - 2.4$ $= 3.8$ m
(d)(i)	$T = kx$ where k is a constant **or** $T = (EA / L)x$ where A is (cross-sectional) area, E is Young modulus, L is (original/unstretched) length
(d)(ii)	$E = \frac{1}{2}kx^2$ **or** $E = \frac{1}{2}Fx$ **and** $F = kx$
	$E = 0.65 \times 2^2$ $= 2.6$ J
	or
	$E = \frac{1}{2}Fx$ $0.65 = \frac{1}{2} \times 270 \times x$ and so $x = 4.8 \times 10^{-3}$ m $k = F / x = 270 / 4.8 \times 10^{-3}$ $= 5.6 \times 10^4$

(d)(ii)	final $E = \frac{1}{2} \times 540 \times 9.6 \times 10^{-3}$ **or** $E = \frac{1}{2} \times (270 \times 2) \times (4.8 \times 10^{-3} \times 2)$ **or** $E = \frac{1}{2} \times 5.6 \times 10^4 \times (9.6 \times 10^{-3})^2$ $= 2.6$ J

36.

(a)	<u>sum</u>/<u>total</u> momentum before = <u>sum</u>/<u>total</u> momentum after **or** <u>sum</u>/<u>total</u> momentum (of a system of objects) is constant
	<u>if no (resultant) external force/for a closed system</u>
(b)(i)	$(3.0 \times 4.0 \times \cos\theta)$ **or** $(2.5 \times 4.8 \times \cos\theta)$ **or** (5.5×3.7)
	$(3.0 \times 4.0 \times \cos\theta) + (2.5 \times 4.8 \times \cos\theta) = (5.5 \times 3.7)$
	$\theta = 32°$
(b)(ii)	(initial $E_K = \frac{1}{2} \times 3.0 \times 4.0^2 + \frac{1}{2} \times 2.5 \times 4.8^2 =$) 53 (J) **or** (final $E_K = \frac{1}{2} \times 5.5 \times 3.7^2 =$) 38 (J)
	values of initial E_K and final E_K both correct **and** inelastic stated

37.

(a)	(velocity =) change in displacement / time
(b)(i)	$a = 2400 / 1200$ $= 2.0$ m s^{-2}
(b)(ii)	straight line from the origin with positive gradient (labelled A)
	ending at (20, 40)
(c)(i)	line starting at origin (with the same gradient as A) and beneath A at all points
	gradient decreasing to zero
	straight horizontal line from t = 12 s and ending at t = 20 s (and labelled B)
(c)(ii)	the velocity/speed will increase
	to a new terminal/constant/maximum velocity/speed
	or
	the car has an acceleration
	to a new (higher) terminal/constant/maximum velocity/speed

38.

(a)	$F = \rho g V$ $0.071 = 1.2 \times 9.81 \times V$
	$(V = 6.03 \times 10^{-3}\ \text{m}^3)$ $4/3 \times \pi \times r^3 = 6.03 \times 10^{-3}$ $r = 0.11$ m
(b)	$m = 0.053 / 9.81$ $(= 5.4 \times 10^{-3}$ kg)
	$F = 0.071 - 0.053$ (= 0.018 N)
	$a = (0.071 - 0.053) / (0.053 / 9.81)$ $= 3.3$ m s^{-2}
(c)(i)	$v^2 = u^2 + 2as$ $3.6^2 = (-1.4)^2 + 2 \times 9.81 \times s$ **or** $3.6^2 = 1.4^2 + 2 \times 9.81 \times s$
	$s = 0.56$ m
(c)(ii)	single straight line from any positive non-zero value of v at $t = 0$ to any negative non-zero value of v at $t = T$
	line starting at (0, 1.4) and ending at (T, –3.6)

39.

(a)(i)	(e.g. a =) [16.5 – 0] / [0.30 – 0] = 55 (m s^{-2})
(a)(ii)	(F =) $T - W$
(a)(iii)	$ma = T - mg$ $m = 16 / (55 + 9.81)$
	$m = 0.25$ kg
(b)(i)	$s = ut + ½at^2$ **or** $v^2 = u^2 + 2as$ **or** $s = vt - ½at^2$ **or** $s = ½(u + v)t$ **or** s = area under graph $s = ½ \times 55 \times 0.30^2$ **or** $s = 16.5^2 / (2 \times 55)$ **or** $s = 16.5 \times 0.30 - ½ \times 55 \times 0.30^2$ **or** $s = ½ \times 16.5 \times 0.30$
	$s = 2.5$ m
(b)(ii)	$u = 16.5$ (m s^{-1})
	$v^2 = u^2 + 2as$ $v^2 = 16.5^2 + 2 \times 9.81 \times 2.5$
	$v = 18$ m s^{-1}

40.

(a)	$E = \frac{1}{2}mv^2$
	$p = mv$
	$m = 0.37^2 / (2 \times 0.30)$ **or** $0.37 / 1.6$ **or** $(0.30 \times 2) / 1.6^2$ $= 0.23$ kg
(b)	$0.37 - 0.65 = -0.13 - p$ $p = 0.15$ kg m s^{-1}
(c)	$7.7 = (0.13 + 0.37) / (\Delta)t$ **or** $7.7 = (0.65 - 0.15) / (\Delta)t$
	time = 0.065 s

41.

(a)	(resultant) force (on an object) is proportional to / equal to the rate of change of momentum
(b)(i)	resultant force = e.g. 6.0 / 4.0 = 1.5 N
(b)(ii)	force X = 1.5 + 2.0 = 3.5 N
(c)	from $t = 0$ to $t = 4.0$ s: horizontal line at any non-zero value of X
	from $t = 0$ to $t = 4.0$ s: horizontal line at $X = 3.5$ N
	from $t = 4.0$ s to $t = 6.0$ s: horizontal line at $X = 2.0$ N

42.

(a)(i)	rate of change of momentum
(a)(ii)	change in momentum = $(1.4 - 0.80) \times 3.0$
	= 1.8 kg m s^{-1}
(b)(i)	resultant force (on block) is zero
	(so) velocity is constant
(b)(ii)	$P = Fv$ **or** $P = Fs / t$
	$v = 2.0 / 0.80$ (= 2.5 m s^{-1})
	distance = 2.5×3.0 = 7.5 m
	or

	$P = W / t$ **or** $P = Fs / t$
	$W = 2.0 \times 3.0$ (= 6.0 J)
	distance = 6.0 / 0.80 = 7.5 m
(c)	0 to 3.0 s: upward sloping straight line from the origin.
	3.0 to 6.0 s: horizontal line at non-zero value of momentum with no 'step change' in momentum at 3.0 s

43.

(a)	$F = 1050 \times 9.81 \times 0.21V$ **or** $W = \rho \times 9.81 \times V$
	$0.21V \times 1050\ (\times 9.81) = V\ (\times 9.81) \times \rho$ $\rho = 220\ \text{kg m}^{-3}$
(b)(i)	$F = 1050 \times 9.81 \times V$ **or** $W = 220 \times 9.81 \times V$
	$(V \times 1050 \times 9.81) - (V \times 220 \times 9.81) = (V \times 220) \times a$
	$a = 37\ \text{m s}^{-2}$
(b)(ii)	the (downward) drag / viscous force increases (with speed)
	resultant force decreases (as upthrust and weight remain the same)
	acceleration decreases (as its velocity increases)

44.

(a)	<u>sum</u> / <u>total</u> momentum before = <u>sum</u> / <u>total</u> momentum after **or** <u>sum</u> / <u>total</u> momentum (of a system of objects) is constant
	if no (resultant) external force / for an isolated system
(b)(i)	$3m \times 4 = m \times v \sin\theta$ ($v \sin\theta = 12$)
	$2m \times 6 = m \times v \cos\theta$ ($v \cos\theta = 12$)
	therefore $\sin\theta = \cos\theta$ **or** $\tan\theta = 1$ $\theta = 45°$
(b)(ii)	$mv \times \cos 45° = 12m$ **or** $mv \times \sin 45° = 12m$ **or** $(mv)^2 = (3m \times 4)^2 + (2m \times 6)^2$
	$v = 17\ \text{m s}^{-1}$
(c)(i)	(chemical energy) = 0.0050 × 700 = 3.5 (J) **or** (chemical energy) = 5.0 × 0.700 = 3.5 (J)

(c)(ii)	$E = \frac{1}{2}mv^2$
	total $E = (0.5 \times 3m \times 4^2) + (0.5 \times 2m \times 6^2) + (0.5 \times m \times 17^2)$
	$3.5 = 204m$ $m = 0.017$ kg

45.

(a)(i)	arrow upwards (↑) **and** labelled upthrust / U arrow downwards (↓) **and** labelled weight / W / mg arrow downwards (↓) **and** labelled tension / T *1 mark: One or two correctly labelled arrows* *2 marks: Three correctly labelled arrows*
(a)(ii)	$U = T + W$ **or** upthrust = tension + weight
	$\rho Vg = T + W$ $V = [(4.00 \times 10^2) + (3.39 \times 10^4)] / (1.23 \times 9.81)$
	$V = 2.84 \times 10^3$ m^3
(a)(iii)	$m = W / g$ **or** $a = F / m$
	$a = (4.00 \times 10^2) / [(3.39 \times 10^4) / 9.81)]$
	$a = 0.12$ m s^{-2}
(b)	there is air resistance (which increases with speed)
	(average) resultant force is less (than weight)
	(average) acceleration is less (than g / 9.81, so speed is less than 100 m s^{-1})
(c)(i)	(extension =) $2.5 \times 2.4 \times 10^{-5} = 6.0 \times 10^{-5}$ (m)
(c)(ii)	$E_{(P)} = \frac{1}{2} Fx$ **or** $E_{(P)} = \frac{1}{2} kx^2$ **and** $F = kx$
	$E_{(P)} = \frac{1}{2} \times 4.00 \times 10^2 \times 6.0 \times 10^{-5}$ **or** $E_{(P)} = \frac{1}{2} \times 6.7 \times 10^6 \times (6.0 \times 10^{-5})^2$ $E_{(P)} = 0.012$ J
(c)(iii)	longer extension **or** smaller spring constant
	elastic potential energy is greater

46.

(a)	displacement
(b)(i)	a = gradient **or** $a = \Delta v / (\Delta)t$ **or** $a = (v - u) / t$
	e.g. $a = (0.30 - 0.12) / (0.35 - 0.15)$ $a = 0.90\ \text{m s}^{-2}$
(b)(ii)	$(0.25 \times 0.48) + (0.75 \times 0.12) = (0.25\,v) + (0.75 \times 0.30)$ **or** $(0.48 - 0.12) = (0.30 - v)$ **or** $(\frac{1}{2} \times 0.25 \times 0.48^2) + (\frac{1}{2} \times 0.75 \times 0.12^2) = (\frac{1}{2} \times 0.25 \times v^2) + (\frac{1}{2} \times 0.75 \times 0.30^2)$
	$v = (-)0.060\ \text{m s}^{-1}$
	direction: to the left / from the right / opposite to (its) initial velocity / opposite to (initial / final) velocity of B
(c)	sketch: horizontal line from (0, 0.48) to (0.15, 0.48)
	horizontal line from (0.35, –0.06) to (0.5, –0.06)
	straight line between (0.15, 0.48) and (0.35, –0.06)

47.

(a)	$65 = \rho g V$ **or** $65 = mg$ **and** $m = \rho V$
	$\rho = 65 / (9.81 \times 7.5)$ $= 0.88\ \text{kg m}^{-3}$
(b)(i)	acceleration = $9.8\ \text{m s}^{-2}$
(b)(ii)	air resistance (acting on sphere) increases
	resultant force (on sphere) decreases
	(magnitude of) acceleration decreases
(c)	$F = ma$ $= 4.0 \times 1.9$ $(= 7.6\ \text{N})$
	resistive force = $(4.0 \times 9.81) - (4.0 \times 1.9)$ $= 32\ \text{N}$

48.

(a)(i)	component of momentum $= 0.25 \times 3.7 \times \sin 27°$
	$= 0.42$ N s
(a)(ii)	$m_{(Z)} \times 5.5 \times \sin 44° = 0.42$ **or** $m_{(Z)} \times 5.5 \times \sin 44° = 0.25 \times 3.7 \times \sin 27°$ $m_Z = 0.11$ kg
(a)(iii)	magnitudes: equal
	directions: opposite
(b)	$4 + 6 = 2 + v$ $v = 8$ m s^{-1}

49.

(a)	resultant force (in any direction) is zero
	resultant moment / torque (about any point) is zero
(b)(i)	$F = \rho V g$ $V = 93\,000 / (1.2 \times 9.81)$ $= 7900$ m^3
(b)(ii)	weight $= 93\,000 + 3(.0) \times 10^3$
	$m = (93\,000 + 3.0 \times 10^3) / 9.81$ $= 9800$ kg
(c)(i)	$(\Delta p) = F(\Delta)t$ **or** $F = \Delta p / (\Delta)t$ **or** F = rate of change of momentum **or** $F = m(v - u)/t$
	$\Delta p = 2800 \times 0.50$ $= 1400$ kg m s^{-1}
(c)(ii)	$(\Delta p =)\ mv$
	$v = 1400 / 64$ $= 22$ m s^{-1}
(c)(iii)	$E_K = ½mv^2$
	$= ½ \times 64 \times 22^2$ $= 15\,000$ J
	or
	$E_K = p^2 / 2m$
	$= 1400^2 / (2 \times 64)$ $= 15\,000$ J